Conditional Diffusion Model With Spatial-Frequency Refinement for SAR-to-Optical Image Translation

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***Abstract*— The presence of speckles and geometric distortions poses a serious challenge to the visual interpretation of synthetic aperture radar (SAR) images. SAR-to-optical (S2O) image translation technology provides a feasible solution and has attracted increasing attention. Restricted by substantial gaps between optical and SAR images, current S2O translation methods unavoidably result in geometric distortions, target missing, and generating low-fidelity images, thereby limiting subsequent cross-modal applications. In this article, we propose an augmented conditional denoising diffusion probabilistic model with spatial-frequency refinement (SFDiff) for high-fidelity S2O image translation. SFDiff progressively narrows the gap between synthesized and real images in both spatial and frequency perspectives, showcasing notable performance in terms of quality and consistency. Specifically, to incorporate rich spatial content priors provided by SAR images, we design an SAR context prior extractor (SCPE) with denoising enhancement to extract multiscale conditional representations, thereby aiding SFDiff in capturing more descriptive cues for S2O translation. In addition, a spatial-frequency complementary learning (SFCL) module is designed to learn spatial semantics and simultaneously enhances informative frequency components and global dependencies. Furthermore, SFDiff is optimized using the joint spatial-frequency refinement loss, facilitating iterative refinement in both spatial and frequency domains to enhance content consistency and fidelity in the synthesized images. Based on the experimental findings from the UNICORN dataset and the SEN12 dataset, SFDiff maintains a high level of content and structural consistency, resulting in visually appealing translation results that surpass the state-of-the-art (SOTA) methods. In particular, SFDiff exhibits excellent performance in preserving small targets and details, which is crucial in cross-modal detection applications.**

***Index Terms*— Conditional diffusion model, cross-modal application, small-target preservation, spatial-frequency refinement, synthetic aperture radar (SAR)-to-optical (S2O) image translation.**

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I. Introduction

BENEFITING from the all-weather all-day operation characteristics, synthetic aperture radar (SAR) has been a crucial remote sensing technology for Earth observation applications [1], [2], [3], [4]. However, due to the coherent imaging mechanism of SAR [5], [6], [7], SAR image response relies on the target's interaction with electromagnetic waves [8], exhibiting undesirable characteristics such as coherent speckles, geometric distortions, and top–bottom inversions. These distortions pose serious spatial misalignment between SAR and optical images, creating substantial challenges for the visual interpretation of SAR data [9]. In contrast, as another mature remote sensing modality, optical remote sensing images (e.g., electro-optical (EO) images) provide intuitive visual representations of ground targets and spatial topology, aligned with human perception [10], [11]. However, as shown in Fig. 1, optical remote sensing imaging relies on the reflection of visible light from surfaces, which is vulnerable to be interfered with clouds, fog, and varying light levels. These factors can significantly degrade the quality of the images captured, often resulting in insufficient visibility and details [12], [13].

One feasible solution to address the aforementioned issues is SAR-to-optical (S2O) image translation. On the one hand, translating SAR images into optical remote sensing images can alleviate the challenge of SAR visual interpretation. On the other hand, when optical remote sensing is affected by bad-weather or low-light conditions [14], [15], converting SAR images to optical images can provide an economical approach to supplement valuable information for cross-modal applications [13]. To be specific, S2O translation allows the use of different pretrained optical image interpretation models on the converted images without redesigning or retraining models for modality fusion. Besides, the translated optical images can also be used for the training data expansion for target detection [8], [16] and spatial–temporal data completion [4], [17]. Over the past few years, S2O translation has become a hot topic in the field, promoting various SAR-optical cross-modal tasks [15], [18], such as SAR-optical matching [19], data fusion [13], [15], and recognition [14], [20], [21], all of which require all-day and all-weather capabilities [11].

Recently, with the enormous development of generative models, a lot of image-to-image (I2I) translation methods [22], [23], [24], [25] have emerged. Prominent examples [22],

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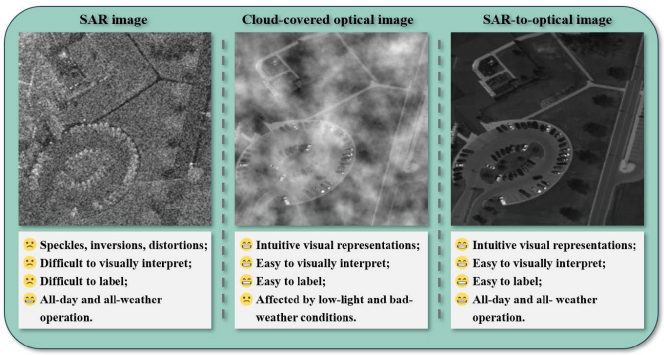


Fig. 1. Illustrating the motivation of S2O translation. When optical image quality is affected by low-light or bad-weather (like cloud) conditions, translating SAR images to optical images can complement the all-weather and all-day operational capabilities of SAR while preserving the visual interpretability advantages of optical images.

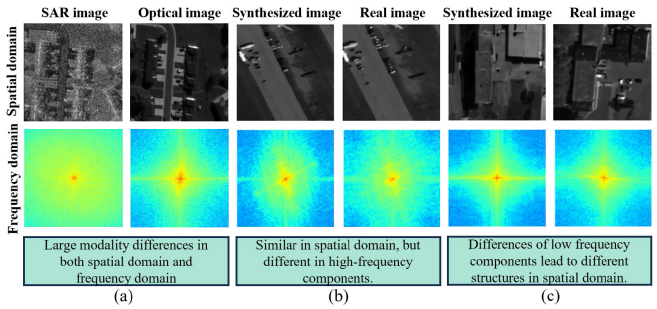


Fig. 2. Illustrating the motivation of joint spatial-frequency refinement. The synthesized images are generated by the baseline model [37]. (a) shows that there are great gaps between the SAR image and the optical image in terms of the frequency domain and the spatial domain. (b) shows that it is easy to observe the high-frequency differences in the frequency domain that are difficult to observe in the spatial domain. (c) shows that low-frequency differences can also imply significant structural differences in the spatial domain. These observations indicate that learning in the frequency domain can provide complementary information (high-frequency components) that is often overlooked in the spatial domain.

[23], [26] employ adversarial learning strategies to perform image translation and have made notable advances. In the context of S2O image translation [14], numerous works [14], [25], [27] leverage adversarial learning strategies to translate SAR images to optical images, augmenting the performance of downstream tasks [16], [18], [28]. Li et al. [1] introduce multiscale wavelet features into generative adversarial networks (GANs) to improve the performance of S2O translation. In [15], researchers use the GAN network to transform optical and SAR images into a common domain to enhance the performance of cross-modal change detection. FG-GAN [14] introduces a fine-grained GAN to alleviate the issue of detail loss in the S2O image translation process.

However, the aforementioned efforts in S2O image translation are mainly based on GANs. Several studies [29], [30] indicate that the adversarial learning strategy is prone to training instability and may lead to mode collapse. This poses a significant challenge for tasks that require strict point-to-point S2O image translation and consequently for many downstream applications [12], [15], [18]. Recent studies [29], [30] have shown that diffusion models have outperformed state-of-the-art (SOTA) GAN methods in terms of the quality and diversity of the synthesized image without suffering from training instability and mode collapse. Unlike GAN-based methods that generate images in a single round [22], diffusion models iteratively add noise according to the Markov chain and optimize the variational lower bound (VLB) to denoise the images over several steps [29]. This paradigm enables diffusion models to more easily learn the data distribution, and it has achieved promising performance in terms of diversity and quality for I2I translation tasks such as text-to-image [31], [32], [33] and image editing [34], [35].

Compared to I2I tasks in natural images, S2O tasks strictly emphasize fidelity in point-to-point image translation [11], which is crucial for downstream tasks such as image registration [13] and target detection [36]. Currently, there are only few works utilizing diffusion models for S2O translation. Bai et al. [10], inspired by diffusion-based super-resolution [37], concatenate SAR images with noise and control diffusion models at the pixel level to generate optical images. However, as shown in Fig. 2(a), SAR images and optical images exhibit significant modality differences in both the spatial and frequency domains. The pixelwise control approach ignores the large modality differences between SAR and optical images, which potentially leads to target loss and geometric distortions during iterative denoising. Furthermore, we find that images generated by diffusion models inevitably have gaps in both spatial and frequency domains. As shown in Fig. 2(b), although the generated images are similar to the real images in the spatial domain, there are significant differences in the high-frequency components according to the frequency spectrum. Moreover, as shown in Fig. 2(c), the differences in the low-frequency components can also imply significant structural differences in the spatial domain. These observations indicate that in the frequency domain, it is possible to capture not only the structural differences (observed in the low-frequency components) but also the high-frequency differences that are difficult to observe in the spatial domain [38]. Furthermore, some researchers have revealed that neural networks prioritize learning low-frequency components while neglecting high-frequency components [39]. Intuitively, leveraging the frequency information effectively can potentially complement the missing informative components and be consequently advantageous for enhancing the quality of S2O images. This intuition is also supported by previous works on image super-resolution [40], [41].

Therefore, we can summarize the aforementioned analysis into the following two main design considerations with respect to the application of diffusion models for S2O translation. On the one hand, an elaborate design for SAR prior control of the diffusion model is crucial to mitigate the effects of modality differences and enables precise control over the S2O process, ensuring ideal performance. On the other hand, the structure and details can be enhanced by learning from the frequency domain. Frequency information can provide complementary information to improve the preservation capability

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TABLE I
STATISTIC COMPARISONS WITH SOTA I2I TRANSLATION METHODS ON THE UNICORN DATASET. THE BOLD FIGURE DENOTES THE BEST SCORE USING THE SYNTHESIZED IMAGES

Method	L2 (↓)	SSIM (↑)	PSNR (↑)	LPIPS (↓)
Pixel2pixel [22]	0.013	0.456	19.11	0.524
Pixel2pixelHD [23]	0.007	0.594	21.28	0.465
SPADE [26]	0.008	0.576	20.87	0.509
ControlNet [31]	0.013	0.493	18.96	0.494
HI-Diff [71]	0.008	**0.654**	21.11	0.524
SR3 [37]	0.013	0.468	18.73	0.462
CDPM [10]	0.009	0.559	20.63	0.49
ResShift [72]	0.007	0.612	21.53	0.438
SFDiff	**0.006**	0.634	**22.30**	**0.385**

TABLE II
STATISTIC COMPARISONS WITH SOTA I2I TRANSLATION METHODS ON THE SEN12 DATASET. THE BOLD FIGURE DENOTES THE BEST SCORE USING THE SYNTHESIZED IMAGES

Method	L2 (↓)	SSIM (↑)	PSNR (↑)	LPIPS (↓)
Pixel2pixel [22]	0.038	0.201	15.24	0.595
SPADE [26]	0.030	0.259	16.10	0.503
ControlNet [31]	0.083	0.151	11.21	0.635
SR3 [37]	0.047	0.385	14.61	0.483
CDPM [10]	0.089	0.312	12.36	0.495
ResShift [72]	0.030	0.332	16.05	0.522
SFDiff	**0.029**	**0.400**	**16.14**	**0.466**

of structure and local details for high-resolution S2O image translation.

Accordingly, with the aforementioned design considerations, this article explores the application of diffusion models in S2O translation and devises an augmented conditional diffusion model with spatial-frequency refinement (SFDiff) for high-fidelity S2O image translation, as shown in Fig. 3. Unlike previous S2O works that control the diffusion model from the pixel level, to reduce the impact of excessive modality differences and achieve precise guidance, the SAR context prior extractor (SCPE) is designed to extract multiscale rich prior knowledge of SAR images and provide hierarchical spatial and semantic controls for the S2O translation. In addition, SFDiff refines the structure consistency and local details from a joint spatial-frequency perspective to improve the fidelity and preservation capability of S2O translation. On the one hand, the spatial-frequency complementary learning (SFCL) module is designed to supplement the missing structures and details by learning informative frequency components and global dependencies [42] in the iterative denoising process. On the other hand, SFDiff introduces the joint spatial-frequency refinement loss to improve the fidelity of generated images by refining the local structural consistency and focal-frequency components. Extensive experiments are conducted on the high-resolution UNICORN dataset [11] and SEN12 dataset [43], and the results demonstrate that SFDiff exhibits significant superiority in both perceptual quality and quantitative metrics compared to the SOTA GAN- and diffusion-based methods.

In summary, the main contributions of this article are given as follows.

1) To reduce the impact of excessive modality differences and achieve effective spatial prior extraction, the SCPE with denoising enhancement is developed to effectively extract the multiscale spatial and semantic priors of SAR images and apply precise hierarchical controls from the feature level for the S2O translation. Experimental results demonstrate that this approach is more effective than the pixel-level control, particularly in cases of significant modality differences.
2) To facilitate precious spatial controls and improve the structural consistency of S2O translation, the SFCL module is designed to complement the missing structures and details during the S2O translation process. By focusing on the joint spatial-frequency perspective, SFCL adaptively completes missing frequency components and global dependencies within spatial features, enhancing the feature representation and improving the preservation capabilities essential for effective S2O translation.
3) To improve the fidelity of S2O translation, the joint spatial-frequency refinement loss is designed to enhance the structural consistency and refine local details from both spatial and spectral perspectives. Through this joint iterative optimization, artifacts caused by the spatial misalignment between SAR and optical images are reduced, resulting in S2O-synthesized images with superior perceptual metrics and greater fidelity.

The remainder of this article is organized as follows. Section II revisits related work. Section III introduces the technological details of SFDiff. Section IV presents experiments and analysis. Section V concludes this article.

II. RELATED WORKS

A. Denoising Diffusion Probabilistic Models

Denoising diffusion probabilistic models (DDPMs), also named diffusion models [29], have emerged as a new family of generative models. Diffusion models generally learn a denoising model to gradually denoise from an original common distribution, e.g., Gaussian noise, to a specific data distribution. DDPM performs a parameterized Markov chain to produce samples of a certain data distribution after a number of steps. In the forward direction, the Markov chain gradually adds noise to the data until it is mapped to a simple isotropic Gaussian distribution. As a consequence of the Markov principle, DDPM [29] exhibits a relatively slower sampling. To address this, various sampling techniques [44], [45], [46] have been developed to enhance the denoising speed. An example is DDIM [44], which introduced a deterministic non-Markovian process to accelerate the sampling process while producing high-quality generations. Leveraging the powerful distribution modeling and generation capabilities, diffusion models have demonstrated promise across various applications within the SAR image community, including target detection [47], [48], [49], SAR image denoising [9], [50], SAR target generation [49], [51], and multimodal fusion [52], [53].

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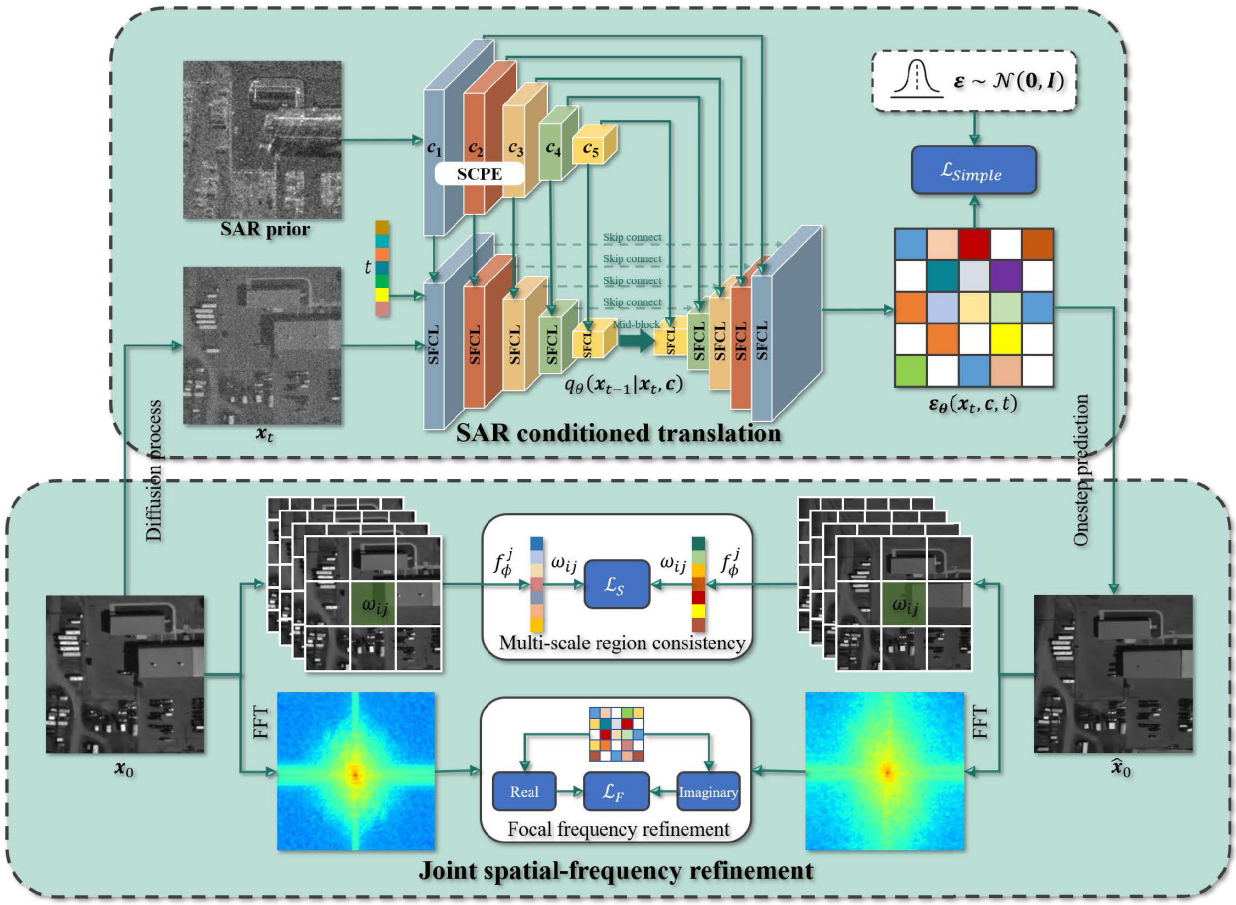


Fig. 3. Overall framework of SFDiff. SCPE is used to extract multiscale SAR priors, and the SFCL module is inserted into the denoising U-Net for hierarchical conditional control of the denoising process. SFDiff is optimized with the joint spatial-frequency refinement loss to improve the quality of synthesized images.

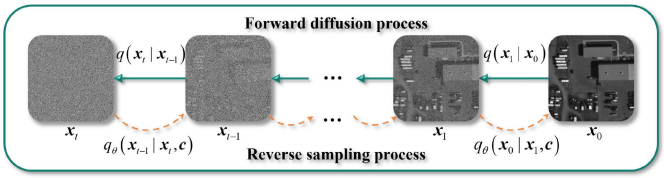


Fig. 4. Illustrating the forward diffusion process and reverse sampling process. In the forward diffusion process, the clear image $\boldsymbol{x}_0$ is gradually diffused by adding the Gaussian noise following the well-defined forward Markovian process q. During the reverse sampling process, the clear image $\boldsymbol{x}_0$ is iteratively denoised from an isotropic Gaussian distribution.

B. S2O Image Translation

Currently, the majority of research on S2O image translation is based on GANs [25], [27], [54], [55], with only a few studies [10] exploring diffusion-based approaches. Some scholars [14], [54], [56] have attempted to construct novel network architectures by using atrous convolutions [56] or multiscale structures [14], [54] to enhance the fidelity of the synthesized optical images. There are also existent methods [27], [57] aiming to improve S2O translation performance by incorporating additional information. In [27], SAR and infrared images are used as auxiliary information to learn discriminative target characteristics and contextual information from coarse to fine levels for S2O translation. Similarly, Guo et al. [58] introduce edge features in GANs to enhance edge preservation ability in the S2O process. Instead of learning from the image level, some studies [1], [55] have also attempted to leverage latent spaces to facilitate S2O image translation. Unlike directly mapping from SAR to optical images, WFLM-GAN [1] maps SAR images to wavelet spaces and reconstructs optical images. In [55], researchers optimize the similarity of hierarchical latent features and employ adjoint reconstruction to achieve S2O translation. Furthermore, by integrating S2O technology, numerous studies [57], [59] have applied S2O translation to enhance the performance of downstream tasks, such as target recognition [60], change detection [18], and cross-modal matching [28]. Regarding the recent popular diffusion models, only a few studies [10] have incorporated them into S2O translation tasks, achieving pixelwise control by concatenating SAR priors with noise. However, due to the speckle noise scattering in SAR images, the pixelwise control may result in suboptimal performance.

III. Methodology

A. Conditional Denoising Diffusion Model

Given the paired data $\{\boldsymbol{c}, \boldsymbol{x}\}$, the conditional denoising diffusion model aims to learn a parametric approximation $p(\boldsymbol{x}|\boldsymbol{c})$ that achieves iterative generation of the target image $\boldsymbol{x}_0$ from a pure noisy image $\boldsymbol{x}_T \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ with a given condition

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TABLE III
ABLATION STUDIES ON DIFFERENT COMPONENTS OF SFDIFF

Methods	SSIM (↑)	LPIPS (↓)
Baseline	0.468	0.462
+SCPE (w/o PPB)	0.551	0.427
+SCPE (PPB)	0.573	0.413
+SCPE +SFCL	0.609	0.401
+SCPE +SFCL +$\mathcal{L}_S$	0.623	0.394
+SCPE +SFCL +$\mathcal{L}_S$ +$\mathcal{L}_F$	**0.634**	**0.385**

$\boldsymbol{c}$. Before that, the conditional diffusion model typically goes through an iterative Gaussian diffusion process [29]. As shown in Fig. 4, given a clear target image $\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0)$, a well-defined forward Markovian diffusion process q is introduced by iteratively adding the Gaussian noise on $\boldsymbol{x}_0$ for T times according to the variance schedule β_t [37]

$$q(\boldsymbol{x}_{1:T}|\boldsymbol{x}_0) = \prod_{t=1}^{T} q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) \tag{1}$$

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I}\right) \tag{2}$$

$\beta_t \in (0, 1)$. According to the linear superposition property of the Gaussian distribution, the noised image $\boldsymbol{x}_t$ can be directly derived as the closed-form solution using (1) and (2)

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0, \left(1-\bar{\alpha}_t\right)\boldsymbol{I}\right) \tag{3}$$

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \tag{4}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$. $\bar{\alpha}_t = \prod_{i=0}^{t}(1-\beta_i)$ indicates the noise level of the tth time step. $t \in (0, T-1)$. If $T \to \infty$, the noised image $\boldsymbol{x}_T$ follows an isotropic Gaussian distribution.

During the reverse process, we can sample from an isotropic Gaussian distribution to progressively reconstruct $\boldsymbol{x}_0$ following the reverse of the well-defined Markovian by approximating the posterior $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{c})$

$$q_{\boldsymbol{\theta}}(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{c}) = \mathcal{N}\left(\boldsymbol{x}_{t-1}; \mu_{\boldsymbol{\theta}}(\boldsymbol{x}_t, \boldsymbol{c}, t), \sigma_t^2 \boldsymbol{I}\right) \tag{5}$$

$$\mu_{\boldsymbol{\theta}}(\boldsymbol{x}_t, \boldsymbol{c}, t) = \frac{1}{\sqrt{1-\beta_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, \boldsymbol{c}, t)\right). \tag{6}$$

Accordingly, given $\boldsymbol{x}_T \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, the original image $\boldsymbol{x}_0$ can be iteratively reconstructed using (5) and (6). Following [29], the $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, \boldsymbol{c}, t)$ can be estimated by a denoising U-Net $\boldsymbol{\theta}$, and the simplified loss function can be formulated as

$$\mathcal{L}_{\text{Simple}} = \mathbb{E}_{\boldsymbol{x}_0, t, \boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon}_{\boldsymbol{\theta}}\left(\sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \boldsymbol{c}, t\right) - \boldsymbol{\epsilon}\right\|^2\right]. \tag{7}$$

B. SAR Context Prior Extractor

The content and structural information in SAR images can provide effective cues for achieving high-fidelity S2O image translation. Previous methods [9], [10] utilize pixelwise control by concatenating noise with SAR images. However, as shown in Fig. 5, the paired optical and SAR targets exhibit significant pixel-level misalignment in radiation characteristics due to differing imaging mechanisms. In addition, the layover phenomenon, caused by the superposition of multiple scatterers in SAR images, has no equivalent in optical images. The shape of shadows in SAR images also differs greatly from those in optical images. These pixel-level misalignments can lead to suboptimal control in the S2O process, particularly in terms of preserving critical structures and details. In contrast, the same targets from different modalities exhibit semantic consistency, while different targets show good separability as shown in Fig. 5. Therefore, as shown in Fig. 6, SFDiff introduces a SCPE module to extract multiscale prior features and inject them hierarchically into the diffusion model using semantic and spatial structural information to guide the S2O translation.

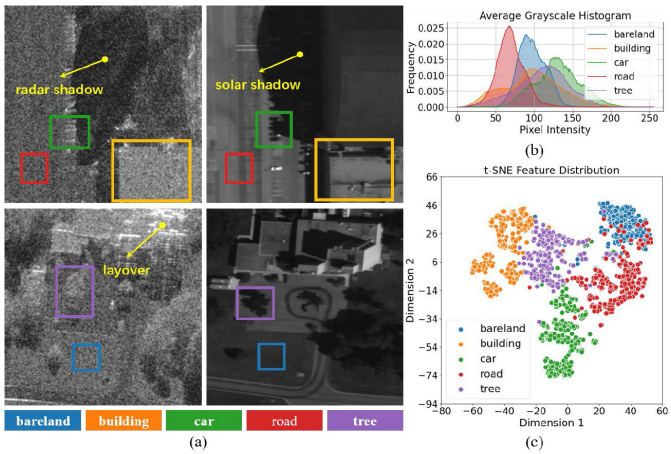


Fig. 5. There are great modality differences (such as geometric distortions, spatial misalignment, and gray-scale range differences) between SAR and optical images, and pixel-level control might not be efficient for S2O image translation. (a) Great geometric and modality differences between different targets. (b) Significant overlaps in the pixel distribution among different SAR targets. (c) Different target classes exhibit semantic distinction in the feature space, while the same class of targets maintains semantic consistency. Therefore, applying semantic control over diffusion at the semantic feature level can mitigate the issue of pixel-level misalignment caused by modality differences.

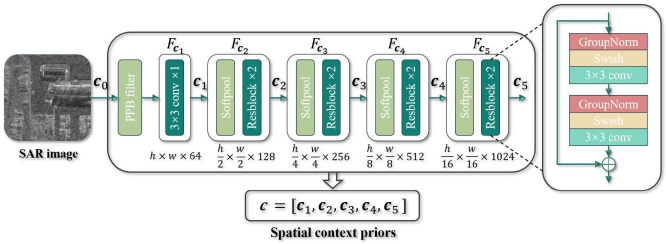
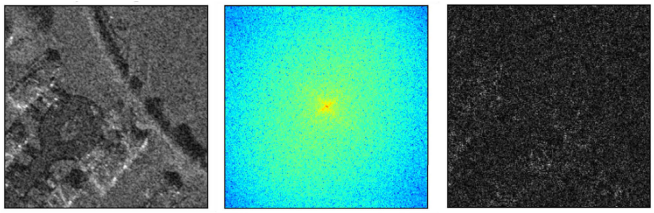


Fig. 6. Structure of the SCPE module.

Before the prior feature extraction, it was observed that the speckle noise in SAR images contains little helpful information for S2O translation. As shown in Fig. 7, the high-frequency components of SAR images mainly consist of noise. Such noise would pose a challenge for prior knowledge extraction [61]. To mitigate the effect of speckles, we adopt a probabilistic patch-based (PPB) filter [61] for high-frequency speckle suppression. As shown in Fig. 8, after suppressing

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(a) (b) (c)

Fig. 7. SAR image spectrum and its high-frequency subband image. (a) Original SAR image. (b) Spectrum of the SAR image. (c) High-frequency subband image.



Fig. 8. Some SAR examples before and after PPB denoising.

the speckle noise, the signal-to-noise ratio (SNR) of the SAR image is improved, resulting in clearer structures and edges in SAR images. Accordingly, a simple yet effective SCPE module F_c is developed to fully exploit informative cues from multiscales in SAR images for S2O translation, as illustrated in Fig. 6. The SCPE module extracts spatial priors from five stages. First, PPB is used to filter the speckles, and a 3 × 3 convolution is followed to extract the texture and surface patterns for the detail control

$$\boldsymbol{c}_1 = \text{Conv}_{3\times3}(\text{PPB}(\boldsymbol{c}_0)). \tag{8}$$

In the following stages, residual blocks are stacked to form the feature extractor $F_{\boldsymbol{c}_i}$ for the deeper semantic prior and spatial structure extraction. In addition, SoftPool [62] instead of conventional pooling is used to mitigate the potential loss of details and control signals for S2O translation

$$\boldsymbol{c}_i = F_{\boldsymbol{c}_i}(\text{SoftPool}(\boldsymbol{c}_{i-1})). \tag{9}$$

Multiscale semantic features $\boldsymbol{c}_i$ are concatenated together to provide feature-level prior controls for the S2O translation

$$\boldsymbol{c} = [\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3, \boldsymbol{c}_4, \boldsymbol{c}_5]. \tag{10}$$

C. SFCL Module

Due to the distinct imaging modalities of SAR and optical images, significant spatial misalignment often occurs, which manifests as differences in the spatial structure and layout of objects with the same semantics. The misalignment can result in suboptimal S2O translation, leading to unavoidable structural and detail loss, and ultimately poor spatial consistency in the generated optical images. To address this issue and enhance structural and detail consistency in the S2O results, a more refined design is necessary to exert precise control

TABLE IV

TARGET DETECTION RESULTS USING THE SYNTHESIZED IMAGES AND THE REAL IMAGES. THE RESULTS ARE PRODUCED BY THE TINY MODEL OF YOLO V3 [73] USING THE TEST DATA. THE BOLD FIGURE DENOTES THE BEST SCORE USING THE SYNTHESIZED IMAGES. THE "REAL IMAGE" ROW SHOWS THE DETECTION RESULTS ON REAL IMAGES

Method	Precision	Recall	F1
ResShift	0.699	0.525	0.601
CDPM	0.762	0.648	0.700
SFDiff	**0.82**	**0.695**	**0.752**
Real image	0.885	0.738	0.805

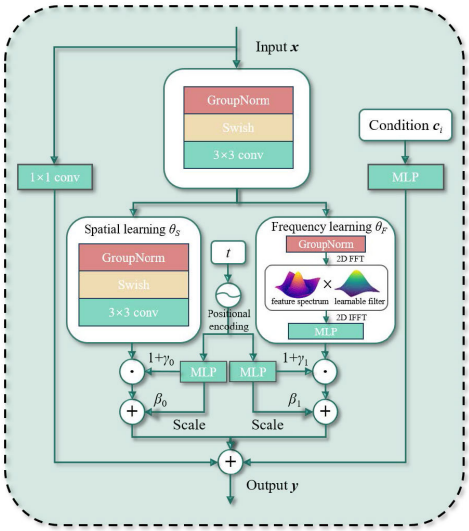


Fig. 9. Structure of the SFCL module. The SFCL module is the basic module that can be stacked to construct the denoising U-Net of SFDiff.

over the generation process and reduce artifacts and detail loss caused by modality discrepancies during S2O translation. To this end, we propose integrating the SFCL module into the denoising U-Net of SFDiff. By leveraging interactions between spatial and frequency contextual features, the SFCL module adaptively compensates for missing frequency components and global dependencies within spatial features, thereby improving the structural integrity and detail consistency of the S2O translation results.

The specific structure of the SFCL module is illustrated in Fig. 9. The SFCL module is a two-branch module, which achieves SFCL. Specifically, given input $\boldsymbol{x}$, the SFCL module produces a spatial-frequency fusion representation $\boldsymbol{y}$ controlled by the SAR prior $\boldsymbol{c}_i$ and modulated by time step t. First, $\boldsymbol{f}_0$ is calculated by a 3 × 3 convolution block according to

$$\boldsymbol{f}_0 = \text{Conv}_{3\times3}(\text{Swish}(\text{GN}(\boldsymbol{x}))) \tag{11}$$

where GN and Swish refer to group normalization [63] and the swish activation function [64], respectively.

Subsequently, the spatial learning branch $\boldsymbol{\theta}_S$ and the frequency learning branch $\boldsymbol{\theta}_F$ are used to extract the spatial representation $\boldsymbol{f}_{\text{spa}}$ and the supplementary frequency feature

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$\boldsymbol{f}_{\text{fre}}$, respectively, by using

$$\boldsymbol{f}_{\text{spa}} = \text{Conv}_{3\times3}\big(\text{Swish}\big(\text{GN}\big(\boldsymbol{f}_0\big)\big)\big) \tag{12}$$

$$\boldsymbol{f}_{\text{fre}} = F_{\text{mlp}}\big(F_{\text{ifft}}\big(F_{\text{fft}}\big(\text{GN}\big(\boldsymbol{f}_0\big)\big)\cdot\mathcal{K}\big)\big) \tag{13}$$

where $\mathcal{K}$ denotes the learnable spectrum kernel. F_{mlp} is a two-layer MLP network.

The spatial feature $\boldsymbol{f}_{\text{spa}}$ and the frequency feature $\boldsymbol{f}_{\text{fre}}$ are then modulated with time step t through the scaling and shifting operations, enabling the SFCL module to perceive noise intensity and to learn spatial-frequency representation during the iterative denoising process

$$\boldsymbol{f}'_{\text{spa}} = \boldsymbol{f}_{\text{spa}}\cdot(1+\gamma_0)+\beta_0 \tag{14}$$

$$\boldsymbol{f}'_{\text{fre}} = \boldsymbol{f}_{\text{fre}}\cdot(1+\gamma_1)+\beta_1 \tag{15}$$

where γ and β are scaling and shifting factors predicted by two independent MLP networks. Before entering the MLP, the time step t is encoded as the sinusoidal positional embedding [65]. Afterward, by aggregating $\boldsymbol{f}'_{\text{spa}}$ with $\boldsymbol{f}'_{\text{fre}}$, the missing frequency components in the spatial features are effectively complemented, resulting in a more enriched and informative combined spatial-frequency representation. The combined representation is then conditioned on the SAR prior feature $\boldsymbol{c}_i$ from the ith scale to learn an effective denoising representation

$$\boldsymbol{f}_1 = \frac{1}{2}\Big(\boldsymbol{f}'_{\text{fre}}+\boldsymbol{f}'_{\text{spa}}\Big)+F_{\text{mlp}}(\boldsymbol{c}_i). \tag{16}$$

Finally, the global residual connection is used to stabilize the feature learning process

$$\boldsymbol{y} = \boldsymbol{f}_1 + \text{Conv}_{1\times1}(\boldsymbol{x}). \tag{17}$$

Like previous works [29], [30], [37], the SFCL module is stacked to construct an encoder–decoder structure, and the denoising U-Net $\boldsymbol{\theta}$ is formed to predict the noise $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, \boldsymbol{c}, t)$ following the reverse Markovian.

D. Joint Spatial-Frequency Refinement

To enhance the structural and local consistency of SFDiff, a region refinement loss and a frequency consistency loss are introduced to jointly refine details in both spatial and frequency domains. Specifically, the spatial region refinement is stimulated by the perceptual consistency of the focal regions at the different scales, while, in the frequency domain, SFDiff is encouraged to complement the missing hard frequency components and enhance the details of the generated images.

1) Region Refinement Loss: Given the synthesized image $\hat{\boldsymbol{x}}_0$ and the real image $\boldsymbol{x}_0$, we aim to maintain perceptual consistency across local regions at different scales $\{2^j\}$. The perceptual distance at the ith region with scale 2^j is represented as

$$d_{ij} = \left\| F_\phi^j\left(\hat{\boldsymbol{x}}_0^{ij}\right) - F_\phi^j\left(\boldsymbol{x}_0^{ij}\right)\right\|_2^2 \tag{18}$$

$j = 2, 3, 4, 5$. Inspired by [66], the nonlinear mapping function F_ϕ^j can be instantiated as a pretrained VGG network [67], which maps local features to the shared perceptual space. The difficulty of learning varies for various local regions. Moreover, it also varies for different time steps t. To take this into account, we formulate the spatial region refinement loss $\mathcal{L}_S$ in a focal manner [68]

$$\mathcal{L}_S = \sum_{ij}(1-t/T)^2 w_{ij}\left\| F_\phi^j\left(\hat{\boldsymbol{x}}_0^{ij}\right) - F_\phi^j\left(\boldsymbol{x}_0^{ij}\right)\right\|_2^2 \tag{19}$$

where $\|\cdot\|_2$ denotes the Euclidean norm. T is the total time step. w_{ij} is the focal reweighing coefficient, which is formulated below to encourage SFDiff to focus on the hard region refinement

$$w_{ij} = \left| F_\phi^j\left(\hat{\boldsymbol{x}}_0^{ij}\right) - F_\phi^j\left(\boldsymbol{x}_0^{ij}\right)\right|. \tag{20}$$

In the implementation, w is further normalized to [0, 1] for the stability.

2) Frequency Consistency Loss: Optimizing the model from the frequency perspective can complement the missing details and correct the spectral distortions that are difficult to observe from the spatial domain, as shown in Fig. 2.

Intuitively, it would be a feasible way to enhance the fidelity of the generated images by maximizing the consistency of the frequency spectrum between the generated and real images. Specifically, the 2-D Fourier spectrum of images at the coordinate (u, v) can be represented as

$$\boldsymbol{F}(u,v) = \boldsymbol{R}(u,v) + j\boldsymbol{I}(u,v) \tag{21}$$

where $\boldsymbol{R}$ and $\boldsymbol{I}$ are the real and imaginary parts, respectively. To more comprehensively characterize the spectral differences, the frequency distance $d(u, v)$ can be formulated by fully considering the complex-valued spectrum

$$\begin{aligned} d(u,v) &= \left\|\boldsymbol{F}(u,v) - \hat{\boldsymbol{F}}(u,v)\right\|_2^2 \\ &= \left(\boldsymbol{R}(u,v) - \hat{\boldsymbol{R}}(u,v)\right)^2 + \left(\boldsymbol{I}(u,v) - \hat{\boldsymbol{I}}(u,v)\right)^2. \end{aligned} \tag{22}$$

Inspired by hard example mining [41], [69], we strive to assign greater emphasis to SFDiff for refining hard frequency components. Consequently, the frequency consistency loss $\mathcal{L}_F$ can be devised by utilizing the focal reweighting manner

$$\mathcal{L}_F = \frac{1}{HW}\sum_{\substack{u\in H\\ v\in W}}\left(1-\frac{t}{T}\right)^2 \bar{w}(u,v)\left\|\boldsymbol{F}(u,v) - \hat{\boldsymbol{F}}(u,v)\right\|_2^2 \tag{23}$$

where H and W are the image height and width, respectively. $\bar{w}(u, v)$ denotes the reweighing coefficient at spectrum coordinate (u, v)

$$\bar{w}(u,v) = \left(\left|\boldsymbol{R}(u,v) - \hat{\boldsymbol{R}}(u,v)\right| + \left|\boldsymbol{I}(u,v) - \hat{\boldsymbol{I}}(u,v)\right|\right)^{k_f} \tag{24}$$

where $k_f = 2$ is the scaling factor. During the training process, $\bar{w}$ is further normalized to [0, 1] for the stability.

3) Joint Training Objective: Consequently, by combining the formulated loss components, the overall training loss function can be expressed as

$$\mathcal{L} = \mathcal{L}_{\text{simple}} + \lambda_1\mathcal{L}_S + \lambda_2\mathcal{L}_F. \tag{25}$$

λ_1 and λ_2 are used for balancing the region refinement loss and the frequency consistency loss, respectively. λ_1 and λ_2 are set to 0.2 and 2 by default in the experiments.

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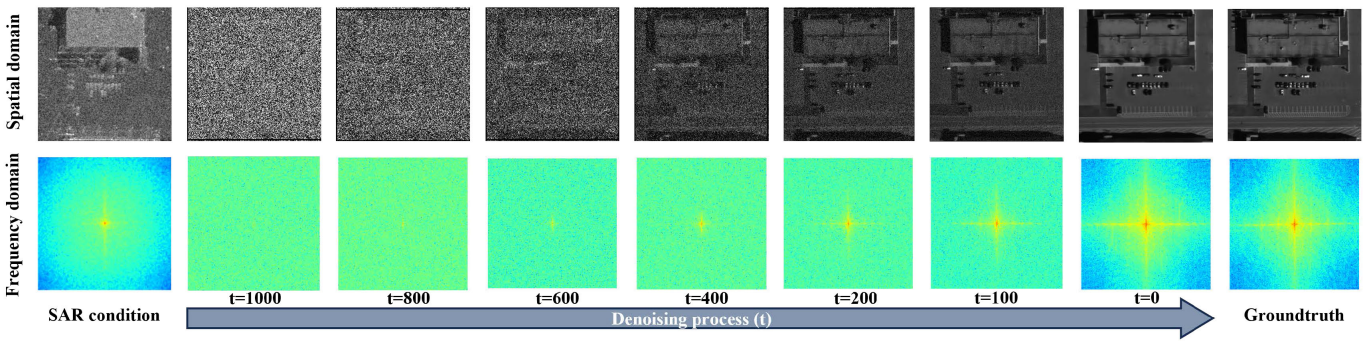


Fig. 10. Illustrating the denoising and refining process of SFDiff. It shows that SFDiff prioritizes synthesizing low-frequency components from noise in the initial stages, gradually refining high-frequency components during the latter half stage of the denoising process.

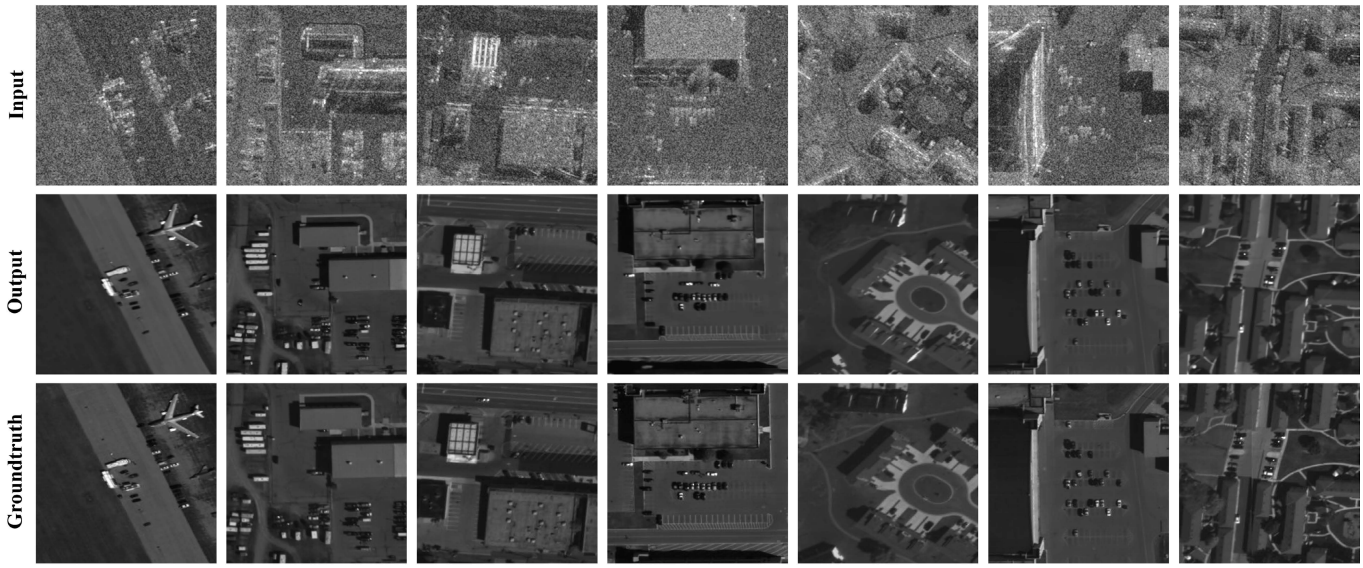


Fig. 11. Partial synthesized images generated by SFDiff. These images show the high local consistency and small-target preservation ability of SFDiff.

IV. Experiments and Analysis

A. Dataset Descriptions

The experimental data are derived from the UNICORN dataset [11], [70] and the SEN12 dataset [43]. The UNICORN dataset is collected by a monochromatic EO sensor and an SAR sensor (HH polarization) in the Wright-Patterson Air Force Base area. There are 68 151 training image pairs and 21 260 validation image pairs in the UNICORN dataset. The resolution is about 0.4 m, and the image size is 256 × 256. Each SAR image contains roughly the same view and matching resolution to the corresponding EO image. The UNICORN dataset contains many small targets, approximately 20 × 20 pixels in size, posing a challenging task for S2O translation. This dataset serves as an effective means to evaluate the model capacity of detail and small-target preservation throughout the S2O translation.

The SEN12 dataset is collected by Sentinel-1 and Sentinel-2 satellites. SAR images are acquired in IW mode with VV polarization of Sentinel-1 C-band radar sensor. The spatial resolution is about 10 m. We obtain 13 600 image pairs for training and 400 image pairs for performance evaluation. Each image is cropped as 256 × 256. The experiment dataset contains a diverse range of scenes, including urban areas, bare land, mountains, water bodies, and farmland, covering various seasons. Such diversity of scenes poses a challenge to the model's adaptability across multiple scenarios.

During the experimental evaluation, L2 metric, structural similarity (SSIM), peak signal-to-noise ratio (PSNR), and learned perceptual image patch similarity (LPIPS) are adopted as metrics to evaluate the quality of the synthesized image. As for the metric calculation, SSIM and PSNR values for both the synthesized image and the ground truth are constrained within the range of 0–255. Conversely, images are normalized within the range of −1 to 1 for the LPIPS metric and for 0 to 1 for the L2 metric.

B. Training and Testing Settings

In the experiments, T is set to 1000. The scheduling of β_t follows a cosine scheduler, increasing from 10^{-6} to 10^{-2} as the time step t increases. The training process of SFDiff is presented in Algorithm 1. The model is trained on a platform with four Quadro RTX6000 24 GB GPUs for about one week. The total batch size is set to 16. The Adam optimizer is employed for training over $N_1 = 600$k iterations for stage 1 and extra 200k iterations for stage 2 until convergence. The

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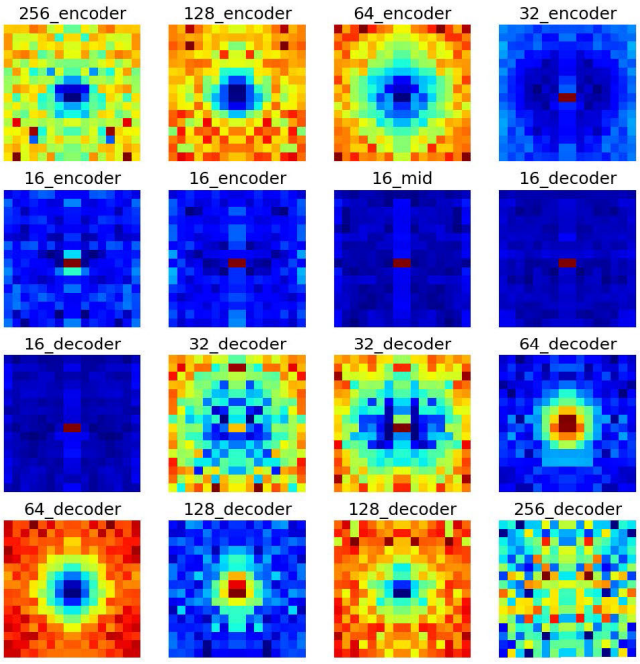


Fig. 12. Visualization of the learned spectrum filter $\mathcal{K}$ is shown. This heatmap is obtained through PCA dimensionality reduction of $\mathcal{K}$, where "n_decoder" denotes that the heatmap is from a decoder layer with a resolution of $n \times n$. Overall, the deeper and intermediate layers tend to supplement low-frequency structures, while the shallow layers tend to enhance high-frequency details.

Algorithm 1 Training Process of SFDiff

Input: paired data $\boldsymbol{c}_0 \sim q(\boldsymbol{c}_0)$ and $\boldsymbol{x}_0 \sim q(\boldsymbol{x}_0)$, U-Net $\boldsymbol{\theta}$, SCPE F_c, and frequency learning branch $\boldsymbol{\theta}_F$.
Output: well-trained U-Net $\boldsymbol{\theta}$ and SCPE F_c.
Initialize iteration N and N_1, freeze $\boldsymbol{\theta}_F$.
for $i = 0$ to N **do**
  Forward diffusion $\boldsymbol{x}_t \leftarrow \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$;
  Extract condition features $\boldsymbol{c} \leftarrow F_c(\boldsymbol{c}_0)$;
  Sample $\hat{\boldsymbol{x}}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(\boldsymbol{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t))$;
  Compute $\mathcal{L}_{\text{Simple}}$ by (7);
  Compute $\mathcal{L}_S$ by $\hat{\boldsymbol{x}}_0$ and (19);
  Compute $\mathcal{L}_F$ by $\hat{\boldsymbol{x}}_0$ and (23);
  Joint loss $\mathcal{L} \leftarrow \mathcal{L}_{simple} + \lambda_1\mathcal{L}_S + \lambda_2\mathcal{L}_F$;
  if $i < N_1$ **then**
    Take gradient descent $\nabla_{\boldsymbol{\theta}}\mathcal{L} + \nabla_{F_c}\mathcal{L}$;
    Update $\boldsymbol{\theta}$ and F_c;
  else
    Unfreeze $\boldsymbol{\theta}_F$;
    Take gradient descent $\nabla_{\boldsymbol{\theta}_F}\mathcal{L}$;
    Update $\boldsymbol{\theta}_F$;
  end if
end for
return $\boldsymbol{\theta}$ and F_c.

initial learning rate is set to 10^{-4}, and it is reduced by ten times at iteration 500k and 700k, respectively. Limited by computational resources, we sample 100 image pairs from the UNICORN dataset and 400 image pairs from the SEN12 dataset, respectively, to evaluate the performance during the testing phase. DDIM with 50 steps is used to accelerate the sampling process in the experiments.

C. Effectiveness Analysis and Comparisons

1) Effectiveness Analysis: Fig. 10 illustrates the spatial-frequency refinement during the sampling process of SFDiff. It shows that as denoising progresses, SFDiff prioritizes synthesizing low-frequency components from noise, gradually refining high-frequency details in the latter half of the denoising process. Quantitative results are shown in Fig. 11, including the S2O results of residential areas, parking lots, and other regions. From Fig. 11, it is evident that SFDiff demonstrates notable image synthesis performance in the S2O translation task. Specifically, the topology of roads, the preservation of small targets, and local textures all achieve high fidelity of visual perception. Besides, it is noteworthy that SFDiff can generate parking lines in parking lot areas, while the parking lines are typically invisible in SAR images. These findings reveal that SFDiff gains the ability to preserve fine details and also highlights its robust contextual understanding and completion capabilities.

2) Comparisons and Analysis: Eight SOTA I2I translation methods are used, including three GAN-based models and five diffusion-based models, to perform the comparison experiments as shown in Tables I and II. We conduct experimental comparisons and analysis from the following aspects:

a) Quantitative comparison: The numerical statistical results are shown in Table I, where SFDiff achieves the optimal performance in terms of PSNR, LPIPS, and L2 score. Regarding the SSIM score, SFDiff is slightly inferior to HI-Diff on UNICORN data. However, as shown in Fig. 12 and 13, by complementing from frequency domain, SFDiff exhibits significantly better visual fidelity compared to HI-Diff from the perspective of perceptual quality, while HI-Diff severely sacrifices high-frequency details. These findings indicate that the SSIM metric prioritizes low-frequency content (i.e., the structure and intensity) in images while neglecting high-frequency details (i.e., textures).

b) Spatial-frequency consistency: From Fig. 13, it can be observed that low-frequency differences lead to significant distortions in the image content and structures, while high-frequency differences result in texture and detail loss. As shown in Fig. 12, SFDiff can learn specific frequency components at different layers, complementing the missing structures and details during the S2O translation process. Accordingly, by comparing the zoomed-in area and the spectrum map in Fig. 13, SFDiff exhibits superior consistency across all frequency ranges, which is supported by better structure consistency (low-frequency) and enhanced detail preservation capability (more complete high-frequency components) in the spatial domain.

c) Target preservation capability: We visualize the comparison results of local details and target preservation, as shown in Fig. 14. We zoom in on the image regions in complex residential areas and parking lots. From these images, it can be seen that the compared methods unavoidably lose vehicle targets, while SFDiff exhibits optimal performance in

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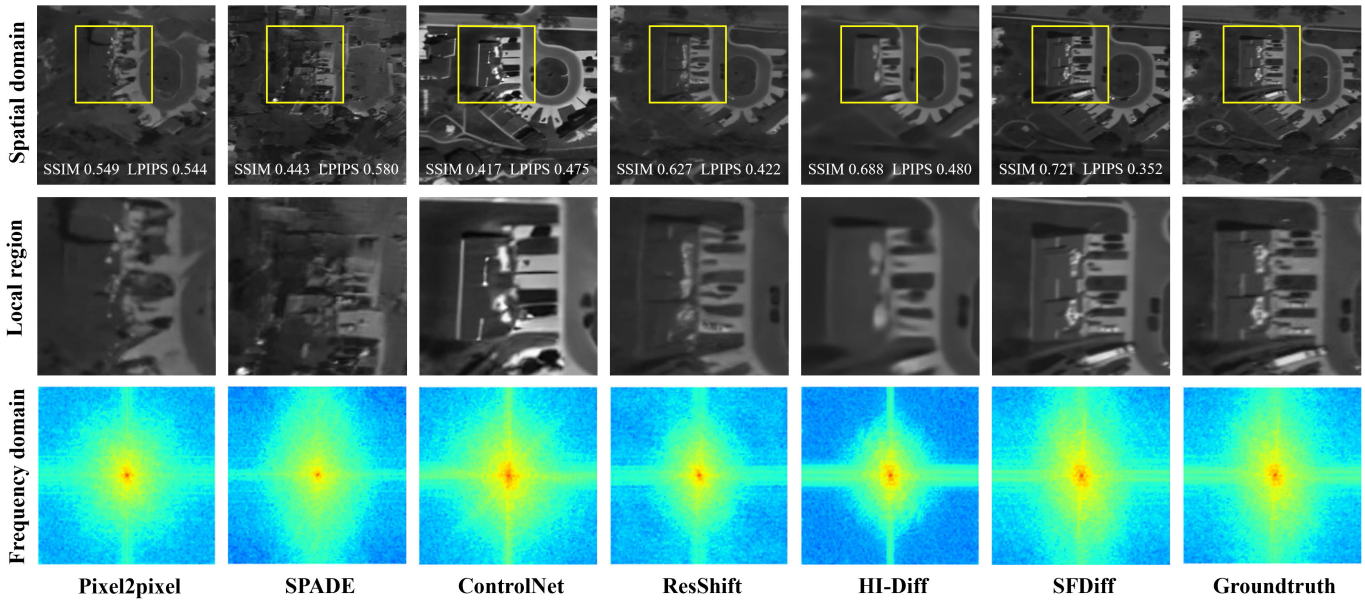


Fig. 13. Qualitative comparisons with SOTA methods from the spatial domain and the frequency domain. The "local region" row denotes the zoomed-in image area of synthesized images.

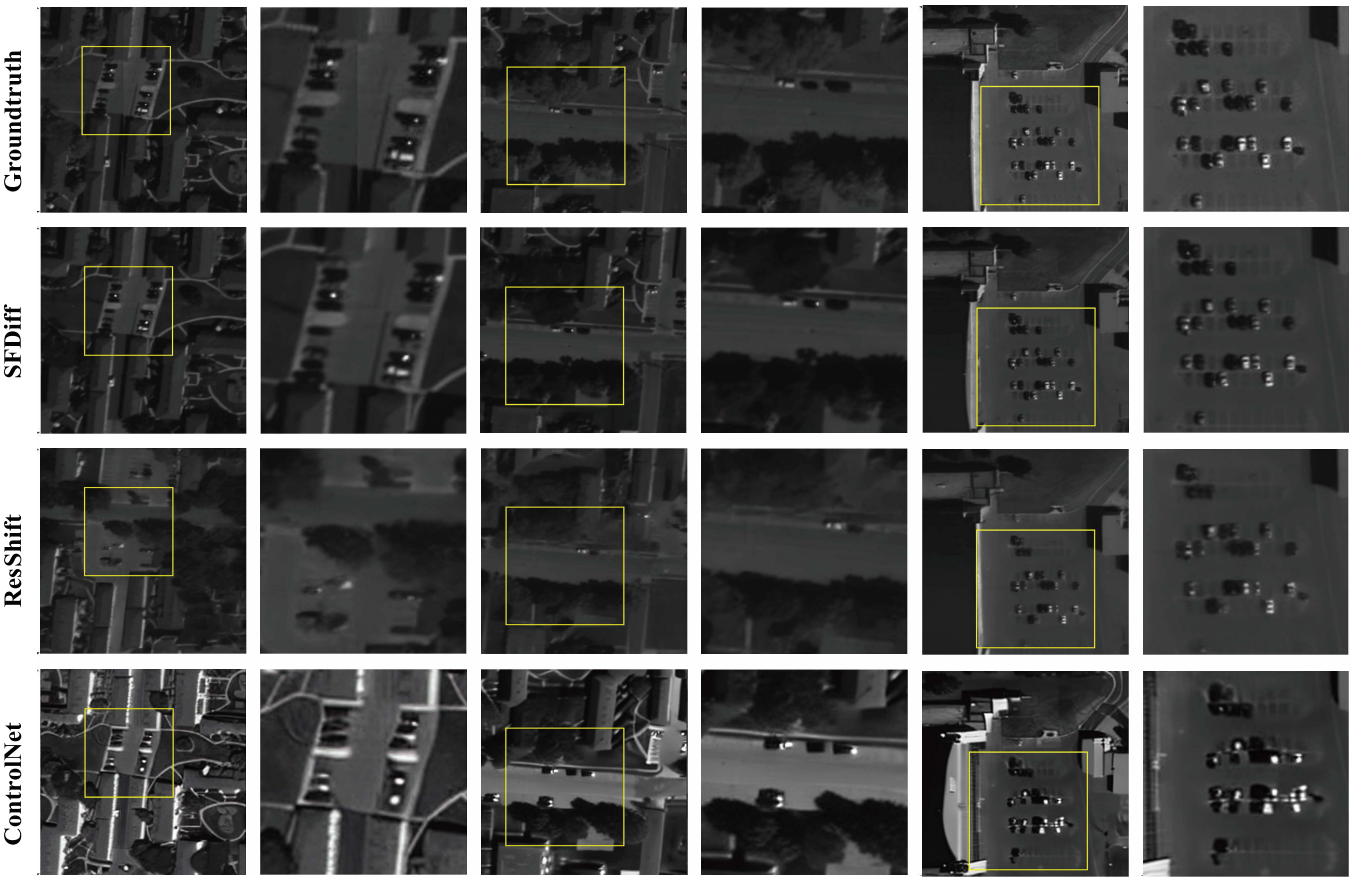


Fig. 14. Illustrating the detail and target preservation ability of different diffusion-based methods. The yellow box and the zoomed-in areas show that other methods inevitably lose some details, even targets in synthesized images.

target preservation and detail representation. These experimental findings reveal that the target preservation ability of SFDiff would exhibit significant potential for SAR-EO cross-modal detection and recognition.

d) Evaluation on the SEN12 dataset: We further conduct performance comparison on the SEN12 dataset, with the results shown in Fig. 15 and Table II. According to the comparison results, SFDiff demonstrates the best performance

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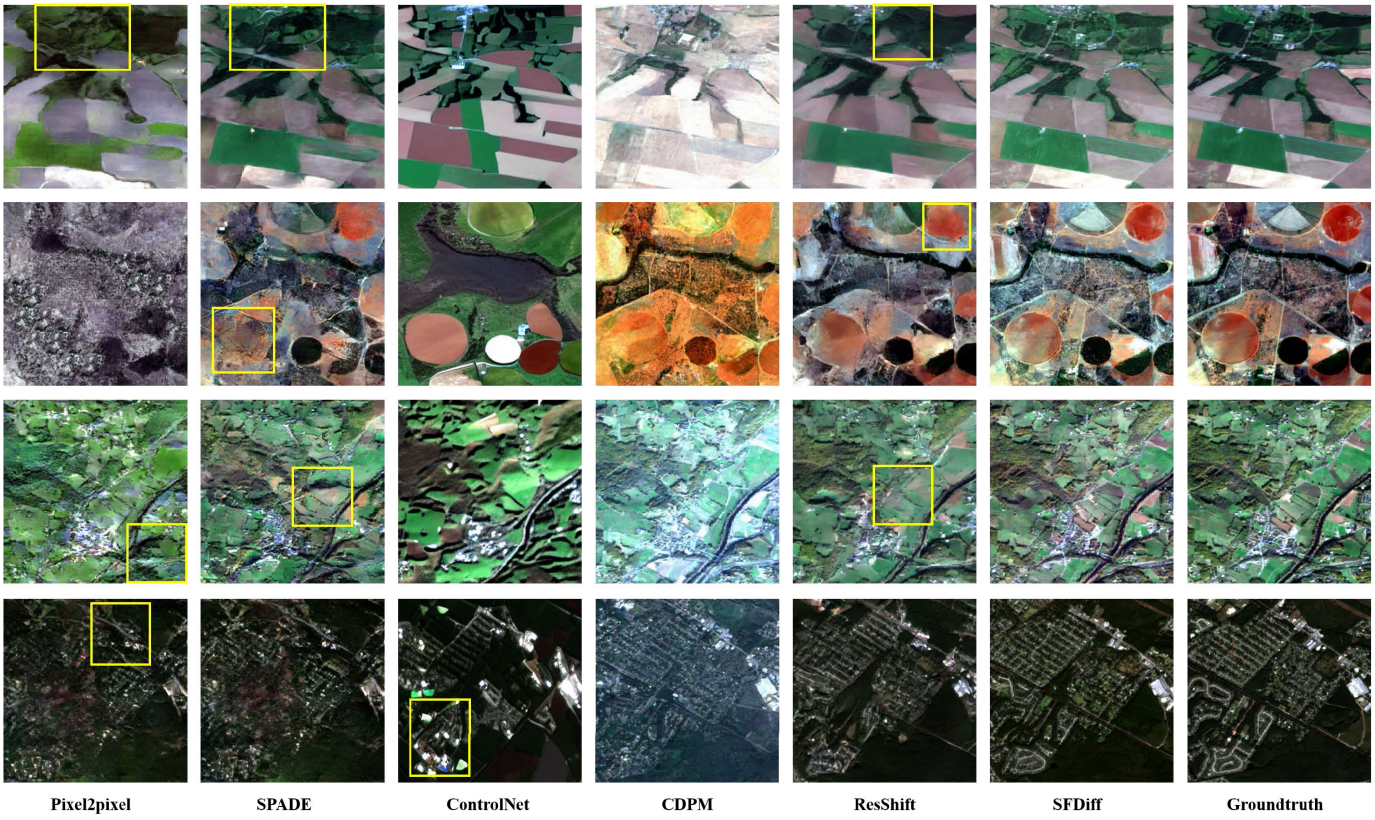


Fig. 15. Qualitative visualization of comparison methods on the SEN12 dataset. The yellow boxes highlight low-fidelity regions.

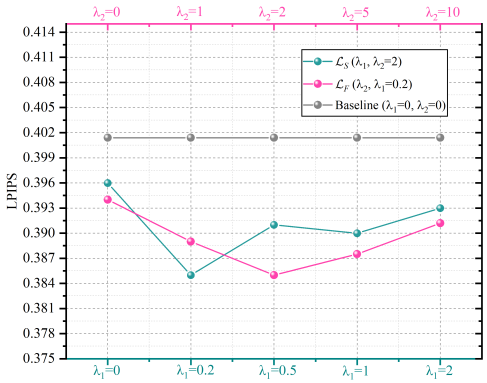


Fig. 16. S2O translation performance with different λ_1's and λ_2's. The LPIPS score is best when $\lambda_1 = 0.2$ and $\lambda_2 = 2$, which are the default settings in our experiments.

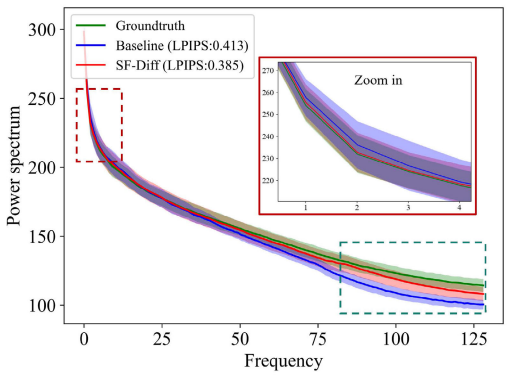


Fig. 17. Average statistics of the 1-D power spectrum using test data. The 1-D power spectrum is computed by azimuthal integration of the Fourier spectrum. Here, "Baseline" denotes SFDiff without the SFCL module $\mathcal{L}_S$ and $\mathcal{L}_F$.

in terms of metric scores and visual quality. As shown in Fig. 15, other diffusion-based models tend to suffer from color distortion (CDPM), information loss (ResShift), and spatial inconsistency (ControlNet). However, for SFDiff, benefiting from the joint spatial-frequency refinement, SFDiff exhibits superior color reconstruction and consistency in structures and details.

D. Ablation Studies and Discussion

1) Ablation Studies: Table III presents the results of the ablation experiments, which are used to evaluate the effectiveness of different components of SFDiff. As SFDiff is built on SR3 [37], SR3 is used as the baseline. The introduction of each module of SFDiff results in varying degrees of improvement in S2O translation performance. In particular, the improvement is most significant when the SCPE module with PPB filtering is introduced, which confirms our previous hypothesis that effectively extracting SAR prior features and applying controls at the feature level are more effective than the pixelwise control. Furthermore, when introducing joint spatial-frequency learning (SFCL module, $\mathcal{L}_S$, and $\mathcal{L}_F$), LPIPS decreases by approximately 3%, and SSIM increases by approximately 6%. This indicates that utilizing joint learning in both the spatial

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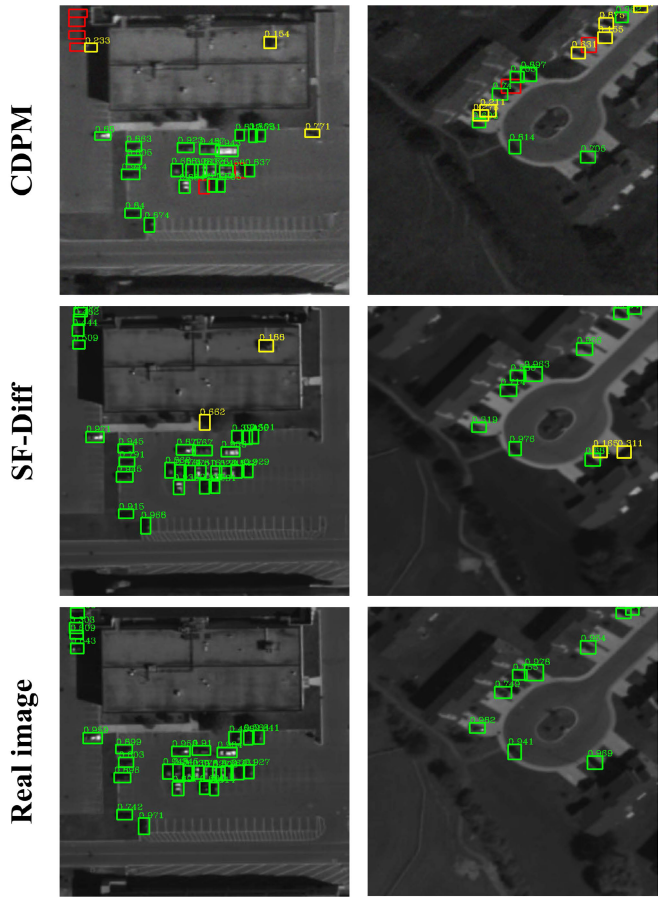


Fig. 18. Target detection results using the real and synthesized images. The red, yellow, and green boxes denote missing targets, false alarms, and detected targets.

and frequency domains can effectively enhance the synthesized image quality of the S2O translation.

2) Discussion on Hyperparameter λ: Fig. 16 illustrates the impact of the balancing weights λ_1 and λ_2 of $\mathcal{L}_S$ and $\mathcal{L}_F$. When $\mathcal{L}_S$ and $\mathcal{L}_F$ are introduced individually in the training process, the LPIPS score of synthesized images becomes better. Moreover, when both $\mathcal{L}_S$ and $\mathcal{L}_F$ are introduced simultaneously, the local perceptual LPIPS score further improves. These findings indicate that joint spatial-frequency refinement contributes to enhancing the quality of S2O images. When $\lambda_1 = 0.2$ and $\lambda_2 = 2$, the visual perceptual quality of SFDiff reaches its optimum.

3) Improvement on Frequency Consistency: Fig. 17 illustrates the differences between synthesized and real images from the frequency domain. Compared to the baseline, SFDiff exhibits better consistency in frequency distribution. Specifically, the images generated by SFDiff are closer to real images in both low-frequency components (as seen in the red zoomed-in area in Fig. 17) and high-frequency components (as seen in the green area in Fig. 17). This indicates that when using spatial-frequency refinement, SFDiff shows varying degrees of improvement in both low- and high-frequency consistency, which also explains why SFDiff has better advantages in visual perception and quantitative metrics.

4) Improvement on Downstream Detection: Table IV and Fig. 18 present the comparative results of target detection using synthesized and real images. The target detection model [73] is trained on real images and tested on synthesized images to assess the fidelity of synthesized images and reveal the potential of SFDiff in downstream cross-modal tasks. Achieving satisfactory cross-modal target detection performance requires synthesized images to closely match the distribution of real images, which places high demands on the target preservation capability of S2O models. Quantitative statistical results from Table IV indicate that compared to ResShift and CDPM, SFDiff achieves higher precision, recall, and $F1$ scores. Visual detection results in Fig. 18 reveal that SFDiff exhibits fewer false alarms and missing targets on synthesized images, which are close to that of real images. These observations reflect that SFDiff not only significantly reduces the distribution gap between synthesized and real images but also has better target preservation capability. The remarkable target detection performance also indicates that the pretrained optical detector can directly perform cross-modal detection on the translated images without retraining or fine-tuning the detection model, providing a promising approach for SAR-assisted cross-modal applications when optical images are ineffective under bad weather conditions.

V. Conclusion

In this article, we introduce an augmented S2O image translation model, termed SFDiff, which is designed to generate high-fidelity optical images through spatial-frequency refinement. SFDiff excels at maintaining content consistency and capturing intricate details by thoroughly incorporating SAR image priors along with a combined approach of spatial-frequency learning. The model utilizes an SCPE to derive SAR image priors effectively and employs an SFCL module to control the feature-level conditions during the denoising process. This dual approach enables SFDiff to significantly enhance the quality of the synthesized images under a spatial-frequency joint optimization framework. Comprehensive evaluations of the UNICORN dataset and the SEN12 dataset have demonstrated the superior performance of SFDiff, showcasing notable improvements over existing SOTA methods in terms of perceptual quality and the preservation of small targets. The capability of SFDiff to maintain these small targets contributes markedly to its performance in downstream target detection tasks, highlighting the potential for cross-modal applications, especially when optical images are rendered ineffective by poor weather conditions. For future work, we plan to extend our exploration of diffusion models within the realm of cross-modal remote sensing applications, aiming to unlock further potential and broaden the applicability of this technology in the field.

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